

Jianjian Gao

POSTDOC RESEARCH ASSOCIATE

School of Data Science, University of Virginia, Charlottesville, Virginia, USA

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🔗 <https://jiaoyang2018.github.io/>

Research Interests

Computational social science; Science of science; Causal inference; Network science

I mainly conduct quantitative research in the field of Science of Science and Computational Social Science using various quantitative methods, including network analysis, causal inference and machine learning.

Academic Experience

2023- **Postdoc research associate**, School of Data Science, University of Virginia.

present Supervisor: Assistant Professor Alexander J. Gates (hgt6rn@virginia.edu)

2017- **Research assistant**, Department of Energy and Power Engineering, Tsinghua University.
2018

Education

Queen Mary University of London

London

PHD IN BUSINESS AND MANAGEMENT (COMPUTATIONAL SOCIAL SCIENCE)

2018 - 2022

- Supervisor: Professor Pietro Panzarasa (p.panzarasa@qmul.ac.uk)

Institute of Engineering Thermophysics, Chinese Academy of Sciences

Beijing

MASTER IN POWER ENGINEERING

2013 - 2016

- Supervisor: Professor Hui Hong; GPA: 84.41/100

Shandong University

Shandong

BACHELOR IN THERMAL ENERGY AND POWER ENGINEERING

2009 - 2013

- GPA: 87.91/100. Outstanding Graduates Awards of Shandong Province (2013); National Endeavour Scholarship (2012); Research and Innovation of Advanced Personal (2011); The First Prize Scholarship (2010); Merit student (2010)

Industry Experience

2016- **Assistant engineer**, Beijing Petrochemical Engineering Co., LTD.(BPEC)
2017

Publications

UNDER REVIEW

Alexander J. Gates, **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science and the diffusion of ideas. <https://arxiv.org/abs/2404.05861> (Nature Communications)

Jianjian Gao, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. (recently submitted)

WORKING PAPERS

Jianjian Gao, Xiancheng Li. Citation Channelling: How large language models reshape reference lists and confound novelty measurement in science.

Jianjian Gao, Alexander J. Gates. The changing landscape of academic leadership: A study of presidents at US R1s.

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. Economic integration and climate change laws.

Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. The diversity of philanthropic boards and the diffusion of scientific policy.

Jianjian Gao, Pietro Panzarasa. Regionalization of international environmental cooperation: Evidence from community structure analysis.

PUBLISHED

Jianjian Gao, Alexander J. Gates. Knowledge Sovereignty: A Path to Independence or Isolation? Research Policy. 2026, 55(8):105557 (<https://doi.org/10.1016/j.respol.2026.105557>)

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. What does network analysis teach us about international environmental cooperation? Ecological Economics. 2023, 205:107670. (One chapter of the PhD thesis)

Stefano Carattini, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. The global network of environmental agreements: a preliminary analysis. Annual Bank Conference on Development Economics 2019, Washington, DC.

Jianjian Gao. International environmental cooperation and climate change laws: A quantitative analysis. 2022. Queen Mary University of London (PhD thesis).

Hui Hong, **Jianjian Gao**, Wanjun Qu, et al. Thermodynamic analyses of the solar-driven Kalina cycle having a variable concentration ratio. Applied Thermal Engineering. 2017, 126: 997-1005. (Part of Master thesis)

Jianjian Gao, Hui Hong, Jie Sun, et al. Study on the performance of concentrated solar Kalina cycle with variable mirror area parabolic trough collector. Journal of Engineering Thermophysics. 2016, 37(8):1595-1601. (EI)

Hao Zhang, Hui Hong, **Jianjian Gao**, et al. Thermodynamic performance of a mid-temperature solar fuel system for cooling, heating and power generation. Applied Thermal Engineering, 2016, 106:1268-1281.

Patents

Hui Hong, Jie Sun, **Jianjian Gao**. Wide-range irradiation and without energy storage concentrating solar energy Kalina power generation system and method. Beijing: CN105156285A, 2015.12.16.

Presentations

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. (Poster)

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. The geography of scientific recognition across disciplines. (Talk)

Sunbelt, June 23-29, 2025, Paris, France – Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. Connected boards, diverse leaders: How networks shape university presidencies. (Talk)

ISSI, June 23-27, 2025, Yerevan, Armenia. **Jianjian Gao**, Alexander J. Gates. Country self-preference and national research systems: A path to independence or isolation? (Talk)

75th Annual ICA Conference, June 12-16, 2025, Denver, Colorado, USA – Gates, Alexander J., **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

ICSSI, Jul 1-3, 2024, DC, USA – **Jianjian Gao** and Gates, Alexander J. The changing landscape of academic leadership: A study of presidents at US R1s. (Poster)

ICSSI, Jul 1-3, 2024, DC, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

IC2S2, Jul 18-20, 2024, Philadelphia, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Poster)

NetSci, Sep 17-25, 2020, Roma, Italy – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. International School and Conference on Network Science. (Poster)

IC2S2, 2020, MA USA – **Jianjian Gao**, Stefano Carattini, Sam Fankhauser, Caterina Gennaioli, Pietro Panzarasa. Structure and Evolution of the International Environmental Cooperation Network. 6th International Conference on Computational Social Science. (Talk)

IC2S2, 2020, MA USA – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. 6th International Conference on Computational Social Science. (Poster)

IC2S2, 2019, Netherlands – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Structure and evolution of the network of countries signing global environmental treaties. 5th International Conference on Computational Social Science. (Poster)

Teaching Experience

Queen Mary University of London

London

TEACHING ASSISTANT

Spring 2022

- **Experiments for business and analytics (BUSM106).**
- This course deals with **advanced econometrics** and **R** programming for Master's students. I mainly taught students how to use **R** to perform all kinds of causal inference, including Difference in Difference (DID), Instrumental Variables (IV), Regression Discontinuity (RD), matching, etc.

Queen Mary University of London

London

TEACHING ASSISTANT

Fall 2022

- **Quantitative Research Methods (BUSM014).**
- This course is designed for Master's students, covering basic elements of **statistical methods**, including descriptive statistics, probability, sampling, inference, and multivariate regression analysis. I mainly taught students how to use **Stata** to perform relevant analysis.

Skills

Programming skills: Python, Stata, R, Bash/shell scripting, HPC, SQL

Computational skills: Network science, Causal inference, Bayesian inference, Machine learning, Deep learning, Natural language processing.

Certificates: Neural Networks and Deep Learning

(<https://coursera.org/share/27bcb57c64bc13975dec99a9be379318>);

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

(<https://coursera.org/share/d790039cd1ce8b299448df4fe8f1fa5a>)

References

Post-doctoral Research Advisor: Assistant Professor Alexander J. Gates

University of Virginia

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<https://alexandergates.net/>

Doctoral Research Advisor: Professor Pietro Panzarasa

Queen Mary University of London
p.panzarasa@qmul.ac.uk
<http://pietropanzarasa.wordpress.com/>

Co-author: Professor Sam Fankhauser

University of Oxford
sam.fankhauser@smithschool.ox.ac.uk
<https://www.ox.ac.uk/news-and-events/find-an-expert/professor-sam-fankhauser>